

DATA ANALYSIS REPORT

Cloudflare Radar 2024 Year in Review

Traffic • Adoption • Disruptions • Security

A data-driven analysis of global Internet trends observed across Cloudflare's network in 2024.

Audience

Technology executives and network infrastructure leaders

April 2026

17% Global Traffic Growth and Starlink's 3.3x Surge

Takeaway: Global traffic grew strongly in 2024; satellite Internet expanded rapidly.



Global Internet traffic

+17%

Year-over-year growth (2024)



Starlink traffic

3.3x

Normalized request volume growth (2024)

Context

Trend line starts mid-January

Normalized for post-holiday activity

Starlink traffic (normalized)



Google Leads Top 10 Services as Generative AI Rises

Takeaway: Google remained #1 overall; generative AI accelerated while the Metaverse faded.

 Top 10 Internet services (Cloudflare 1.1.1.1 DNS)

#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Methodology: Amazon 1.1.1.1 resolver traffic

Figure: Top 10 services worldwide

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI: continued rapid rise throughout 2024
Metaverse receded into the background

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Takeaway: Protocol and crypto adoption shifted materially in 2024, driven by browser defaults.



Post-quantum key agreement

~2% → ~13%

HTTP/3 adoption

21%

Top API client language

Go (~12%)

Post-quantum encrypted traffic (TLS 1.3)



Trigger: Chrome 124 enabled post-quantum by default on 2024-04-17; adoption climbed to ~12–13%.

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Takeaway: Disruptions were frequent and often policy-driven, stressing resiliency planning.



Major disruptions observed

225

Globally during 2024

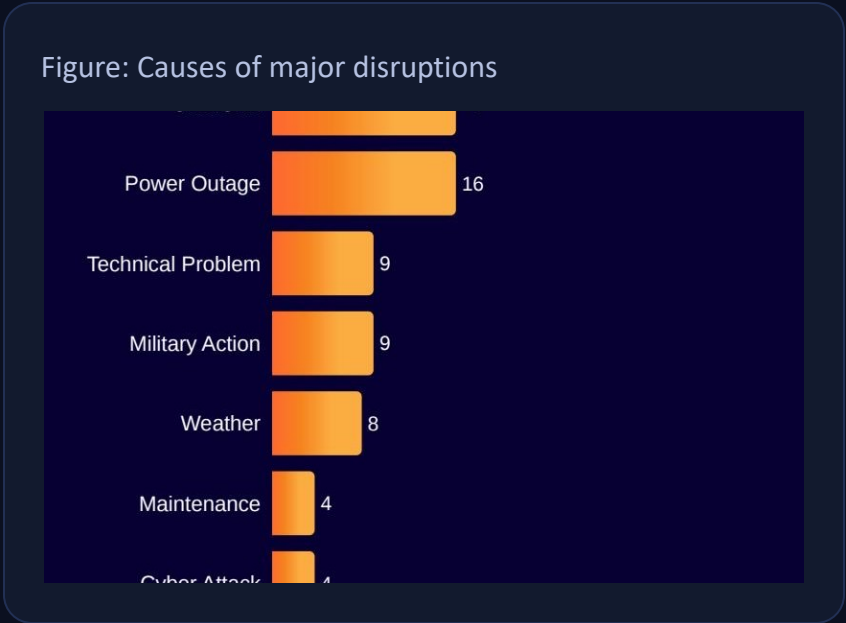
Leading causes (counts)

Government-directed shutdowns (116+), cable cuts (41), power outages (16)

Cloudflare Radar Outage Center tracks scope and duration using traffic signals

Disruptions by cause (2024)

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Takeaway: Threat volume remained high; long-tail vulnerabilities and email deception persist.

 Traffic mitigated
6.5%
Of all traffic processed

Most attacked sector
Gambling & Games
Bot concentration: 68% of global bot traffic originated from the top 10 countries

Persistent exploit: Log4j
CVE-2021-44228 continues to be actively targeted
First disclosed: Dec 2021 — still exploited in 2024

Routing security
RPKI valid routes increased by **6.4%**

Email threats (Cloudflare Email Security)
Malicious emails: **4.3%**
Of malicious emails, those with deceptive links: **43%**
Top malicious/spam originating TLD: **.bar**

Interpretation: email deception remains a consistent initial-access vector.

Five Takeaways Shaping the 2025 Internet Outlook

Takeaway: 2024 signals point to five priorities for 2025 planning.



Strategic takeaways (grounded in 2024 Radar observations)

- 1 Plan for continued traffic growth**
Global Internet traffic grew 17% YoY — capacity and performance investments remain table stakes.
- 2 Accelerate post-quantum readiness**
Post-quantum encryption climbed from ~2% to ~13% of TLS 1.3 traffic after browser defaults changed.
- 3 Strengthen resilience against policy-driven outages**
Government-directed shutdowns were the leading cause among 225 major disruptions observed in 2024.
- 4 Assume persistent, high-volume threats**
Cloudflare mitigated 6.5% of all processed traffic; Log4j exploitation continued years after disclosure.
- 5 Track developer ecosystem shifts**
Go became the top API client language (~12% of automated API requests); HTTP/3 reached 21% adoption.